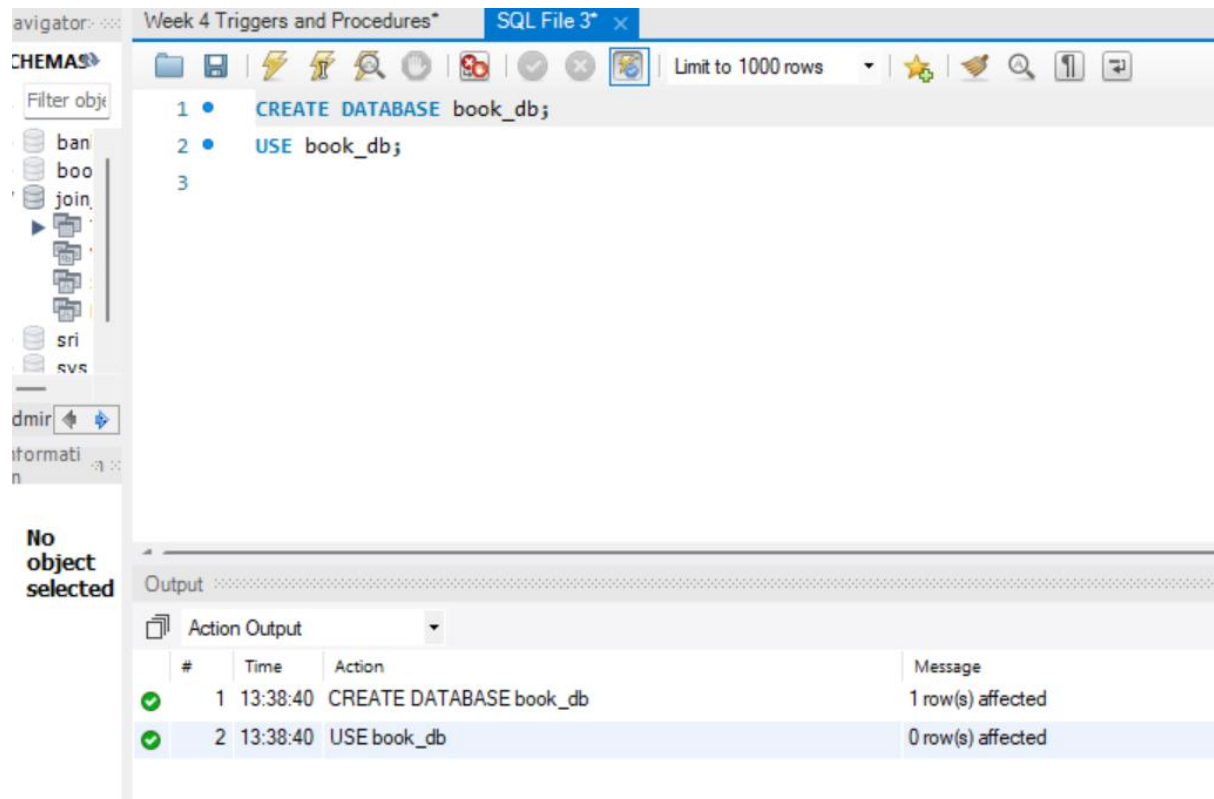


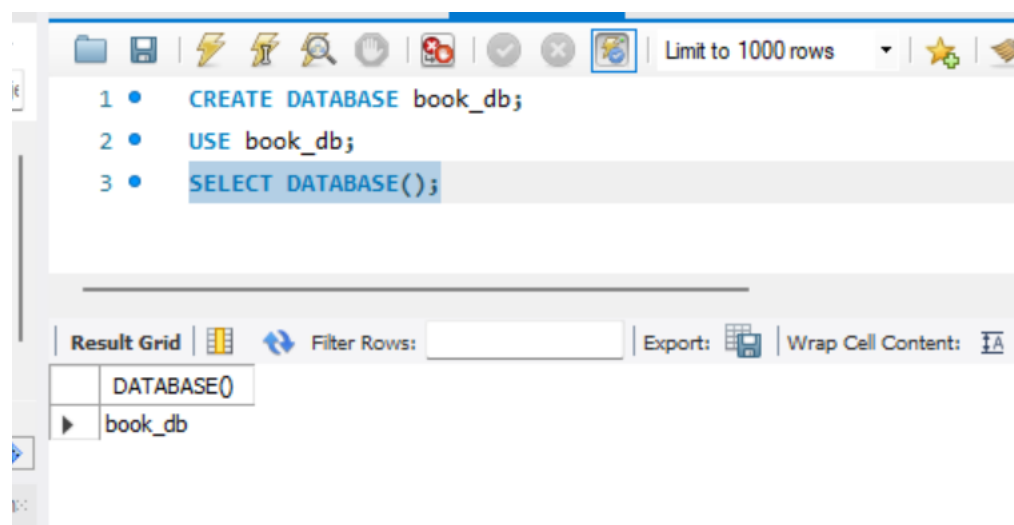
Practical 4

```
CREATE DATABASE bookflow_db;
```

```
USE bookflow_db;
```



```
SELECT DATABASE();
```



```
CREATE TABLE Books
```

```
(
book_id INT AUTO_INCREMENT PRIMARY KEY,

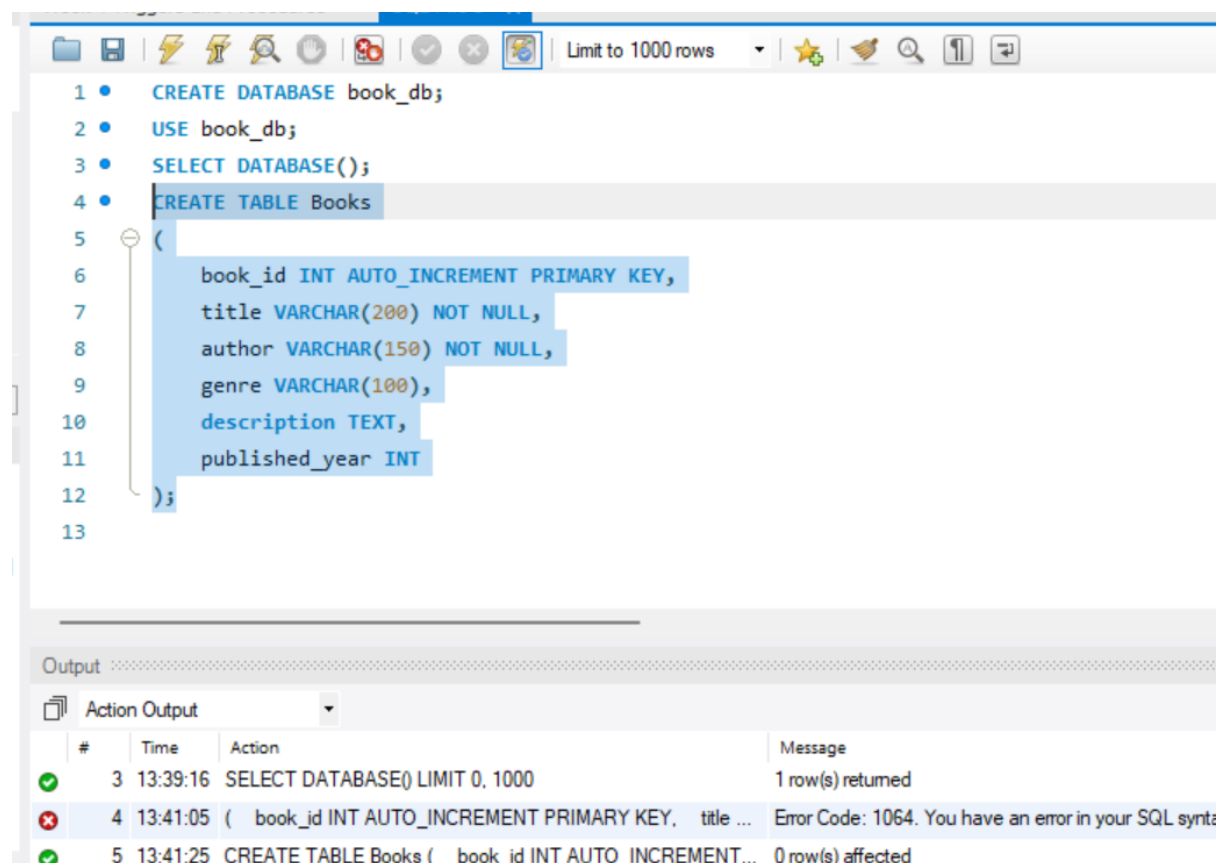
title VARCHAR(200) NOT NULL,

author VARCHAR(150) NOT NULL,

genre VARCHAR(100),

description TEXT,

published_year INT
);
```



The screenshot shows a SQL IDE with a query editor and an output window. The query editor contains the following SQL code:

```
1 • CREATE DATABASE book_db;
2 • USE book_db;
3 • SELECT DATABASE();
4 • CREATE TABLE Books
5 • (
6 •     book_id INT AUTO_INCREMENT PRIMARY KEY,
7 •     title VARCHAR(200) NOT NULL,
8 •     author VARCHAR(150) NOT NULL,
9 •     genre VARCHAR(100),
10 •    description TEXT,
11 •    published_year INT
12 • );
13
```

The output window shows the results of the execution:

#	Time	Action	Message
3	13:39:16	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
4	13:41:05	(book_id INT AUTO_INCREMENT PRIMARY KEY, title ...	Error Code: 1064. You have an error in your SQL synta
5	13:41:25	CREATE TABLE Books (book_id INT AUTO_INCREMENT...	0 row(s) affected

```
DESCRIBE Books;
```

Execute the selected portion of the script or everything, if there is no selection

```

9
10     description TEXT,
11     published_year INT
12 );
13 • DESCRIBE Books;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [↕](#)

	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(200)	NO		NULL	
	author	varchar(150)	NO		NULL	
	genre	varchar(100)	YES		NULL	
	description	text	YES		NULL	
	published_year	int	YES		NULL	

INSERT INTO Books

(

title,

author,

genre,

description,

published_year

)

VALUES

(

'The Martian',

'Andy Weir',

'Science Fiction',

'An astronaut struggles to survive alone on Mars.',

2011

),

(
 'Dune',
 'Frank Herbert',
 'Science Fiction',
 'A story involving planets, politics, survival and space.',
 1965
)

(
 'The Hitchhikers Guide to the Galaxy',
 'Douglas Adams',
 'Science Fiction',
 'A humorous adventure across space and alien worlds.',
 1979
)

(
 'Pride and Prejudice',
 'Jane Austen',
 'Romance',
 'A classic story about relationships and social expectations.',
 1813
)

(
 'The Great Gatsby',
 'F. Scott Fitzgerald',
 'Classic',
 'A story of ambition, wealth, love and the American dream.',
 1925
)

(
 'Foundation',
 'Isaac Asimov',
 'Science Fiction',
 'A futuristic story about mathematics, civilization and space.',
 1951
)

(
 'The Hobbit',
 'J.R.R. Tolkien',
 'Fantasy',
 'A fantasy adventure involving Bilbo Baggins and a dangerous journey.',
 1937
)

(
 'Enders Game',
 'Orson Scott Card',
 'Science Fiction',
 'A young student is trained for an interstellar conflict.',
 1985
)

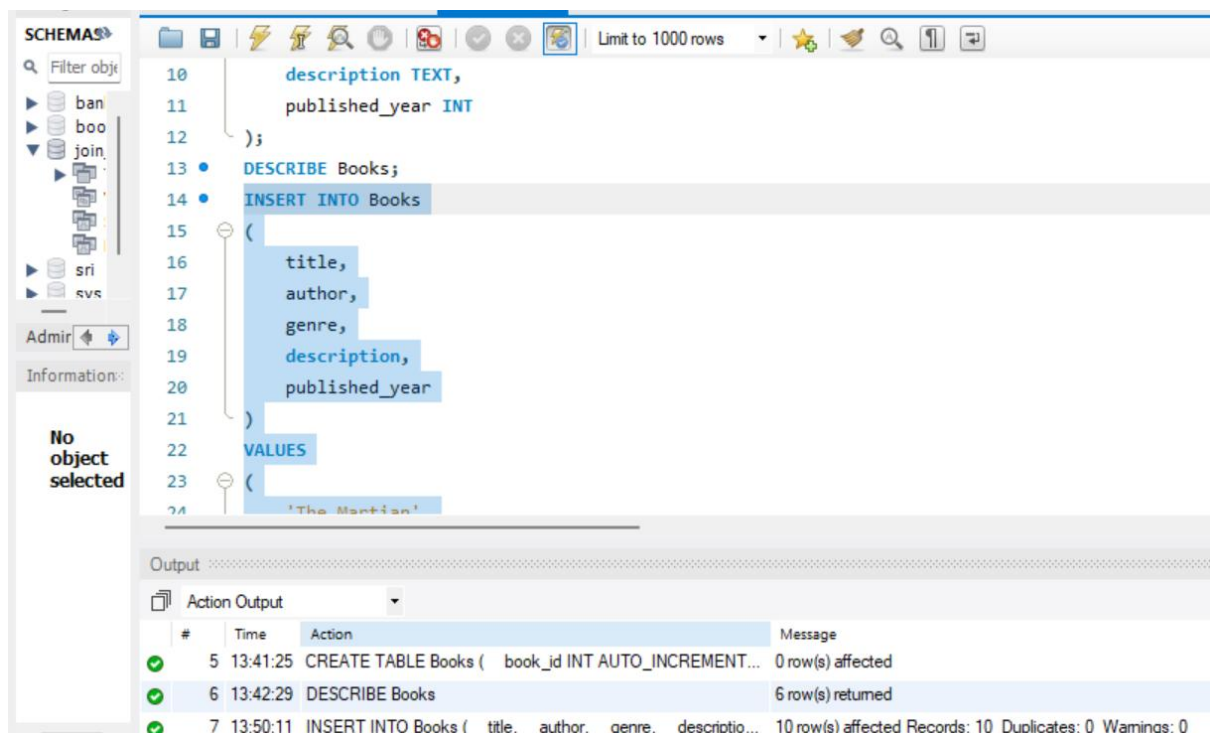
(
 'Harry Potter and the Sorcerers Stone',
 'J.K. Rowling',
 'Fantasy',
 'A young wizard begins his magical education.',
 1997

```

),

(
    'Twenty Thousand Leagues Under the Seas',
    'Jules Verne',
    'Adventure',
    'A fantastic underwater adventure aboard a mysterious submarine.',
    1870
);

```



```

SELECT
    book_id,
    title,
    author,
    genre,
    published_year
FROM Books
ORDER BY book_id;

```

Week 4 Triggers and Procedures* SQL File 3*

```

94     book_id,
95     title,
96     author,
97     genre,
98     published_year
99 FROM Books
100 ORDER BY book_id;

```

Result Grid

	book_id	title	author	genre	published_year
1	1	The Martian	Andy Weir	Science Fiction	2011
2	2	Dune	Dune k Herbert	Science Fiction	1965
3	3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	4	Pride and Prejudice	Jane Austen	Romance	1813
5	5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925

Books 3

Output

Action Output

#	Time	Action	Message	Duration / Fetc
6	13:42:29	DESCRIBE Books	6 row(s) returned	0.000 sec / 0.0
7	13:50:11	INSERT INTO Books (title, author, genre, descriptio...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.015 sec
8	13:51:01	SELECT book_id, title, author, genre, published_y...	10 row(s) returned	0.000 sec / 0.0

ALTER TABLE Books

ADD COLUMN embedding JSON;

96 author,
97 genre,
98 published_year
99 FROM Books
100 ORDER BY book_id;
101 • ALTER TABLE Books
102 ADD COLUMN embedding JSON;
103
104

Output

Action Output

#	Time	Action
7	13:50:11	INSERT INTO Books (title, author, genre, descriptio..
8	13:51:01	SELECT book_id, title, author, genre, published_y..
9	13:51:45	ALTER TABLE Books ADD COLUMN embedding JSON

DESCRIBE Books;

103 • DESCRIBE Books;

Result Grid | Filter Rows: | Export: | Wrap Cell

	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increm
	title	varchar(200)	NO		NULL	
	author	varchar(150)	NO		NULL	
	genre	varchar(100)	YES		NULL	
	description	text	YES		NULL	

Result 4 x

Output

Action Output

	#	Time	Action
✓	8	13:51:01	SELECT book_id, title, author, genre, publishe
✓	9	13:51:45	ALTER TABLE Books ADD COLUMN embedding JSON
✓	10	13:52:28	DESCRIBE Books

UPDATE Books

SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)

WHERE book_id = 1;

Week 4 Triggers and Procedures* SQL File 3*

Execute the selected portion of the script or everything, if there is no selection

```

98
99 FROM Books
100 ORDER BY book_id;
101 • ALTER TABLE Books
102 ADD COLUMN embedding JSON;
103 • DESCRIBE Books;
104 • UPDATE Books
105 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
106 WHERE book_id = 1;
107
108
109
110
111

```

Output

Action Output

#	Time	Action	Message
9	13:51:45	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected
10	13:52:28	DESCRIBE Books	7 row(s) returned
11	13:53:11	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 row(s) affected

UPDATE Books

SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)

WHERE book_id = 2;

```

107 • UPDATE Books
108 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
109 WHERE book_id = 2;
110
111
112

```

Output

Action Output

#	Time	Action	Message
10	13:52:28	DESCRIBE Books	7 row(s) returned
11	13:53:11	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 row(s) affected
12	13:53:48	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 row(s) affected

UPDATE Books

SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)

WHERE book_id = 3;

The screenshot shows a SQL editor with the following code:

```
110 • UPDATE Books
111     SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
112     WHERE book_id = 3;
113
114
```

Below the editor is an "Output" section with a dropdown menu set to "Action Output". It displays a table with the following data:

	#	Time	Action	M
✓	11	13:53:11	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 n
✓	12	13:53:48	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 n
✓	13	13:54:21	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, ...	1 n

UPDATE Books

SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)

WHERE book_id = 4;

The screenshot shows a SQL editor with the following code:

```
113 • UPDATE Books
114     SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
115     WHERE book_id = 4;
116
117
```

Below the editor is an "Output" section with a dropdown menu set to "Action Output". It displays a table with the following data:

	#	Time	Action	Message
✓	12	13:53:48	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 row(s) aff
✓	13	13:54:21	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, ...	1 row(s) aff
✓	14	13:54:49	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, ...	1 row(s) aff

UPDATE Books

SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)

WHERE book_id = 5;

```

115 WHERE book_id = 4;
116 • UPDATE Books
117 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
118 WHERE book_id = 5;
119
120

```

Output

Action Output

#	Time	Action
✓ 13	13:54:21	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70
✓ 14	13:54:49	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20
✓ 15	13:55:15	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30

UPDATE Books

SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)

WHERE book_id = 6;

```

119 • UPDATE Books
120 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
121 WHERE book_id = 6;
122
123

```

Output

Action Output

#	Time	Action	Message
✓ 14	13:54:49	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, ...	1 row(s) affe
✓ 15	13:55:15	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, ...	1 row(s) affe
✓ 16	13:55:52	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, ...	1 row(s) affe

UPDATE Books

SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)

WHERE book_id = 7;

```

UPDATE Books
SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
WHERE book_id = 7;

```

tion Output

	Time	Action	Message
5	13:55:15	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, ...	1 row(s) affected Rows i
6	13:55:52	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, ...	1 row(s) affected Rows i
7	13:56:36	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, ...	1 row(s) affected Rows i

UPDATE Books

SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)

WHERE book_id = 8;

```

125 • UPDATE Books
126 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
127 WHERE book_id = 8;
128
129

```

Output

Action Output

	#	Time	Action
✓	16	13:55:52	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65,
✓	17	13:56:36	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75,
✓	18	13:57:05	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85,

UPDATE Books

SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)

WHERE book_id = 9;

```
128 • UPDATE Books
129 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
130 WHERE book_id = 9;
131
132
```

Output

Action Output

#	Time	Action	M
✓ 17	13:56:36	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, ...	1
✓ 18	13:57:05	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, ...	1
✓ 19	13:57:45	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, ...	1

UPDATE Books

SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)

WHERE book_id = 10;

```
131 • UPDATE Books
132 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
133 WHERE book_id = 10;
134
135
```

Output

Action Output

#	Time	Action	M
✓ 18	13:57:05	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, ...	1
✓ 19	13:57:45	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, ...	1
✓ 20	13:58:31	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, ...	1

SELECT

book_id,

title,

embedding

FROM Books

ORDER BY book_id;

134	•	SELECT
135		book_id,
136		title,
137		embedding
138		FROM Books
139		ORDER BY book_id;
140		

Result Grid	Filter Rows:	Edit:	Exp
book_id	title	embedding	
1	The Martian	[0.90, 0.80, 0.20]	
2	Dune	[0.85, 0.75, 0.25]	
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]	
4	Pride and Prejudice	[0.10, 0.20, 0.90]	
5	The Great Gatsby	[0.15, 0.30, 0.80]	

Books 5 x

SELECT

book_id,

title,

author,

genre,

ROUND(

SQRT(

POW(

CAST(

JSON_EXTRACT(embedding, '\$[0]')

AS DECIMAL(10,6)

) - 0.90,

2

)

+

POW(

CAST(

JSON_EXTRACT(embedding, '\$[1]')

AS DECIMAL(10,6)

) - 0.80,

2

)

+

POW(

CAST(

JSON_EXTRACT(embedding, '\$[2]')

AS DECIMAL(10,6)

) - 0.20,

2

)

),

4

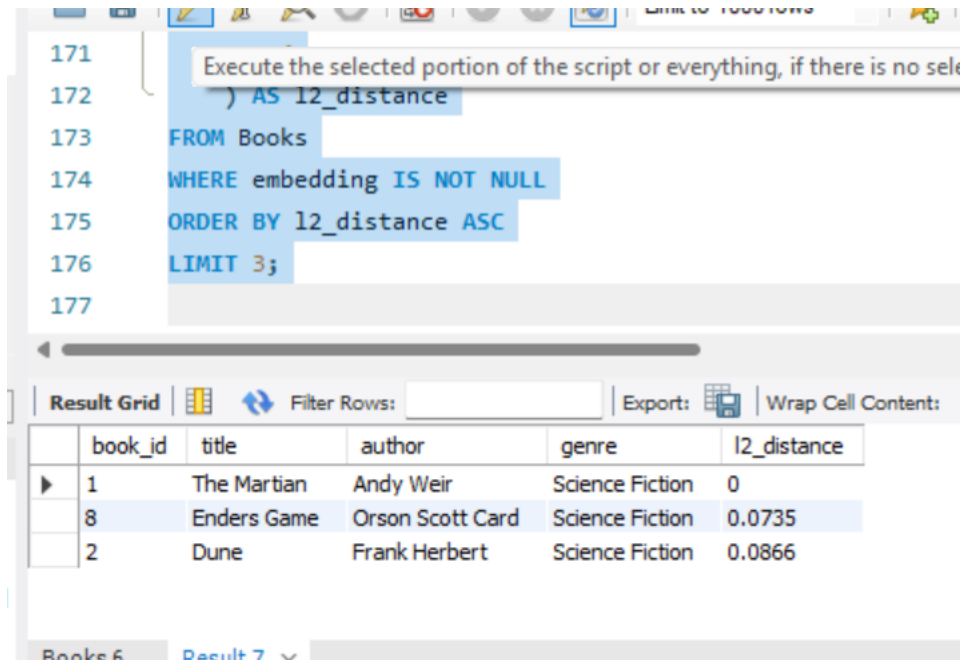
) AS l2_distance

FROM Books

WHERE embedding IS NOT NULL

ORDER BY l2_distance ASC

LIMIT 3;



The screenshot shows a SQL query editor with a query window and a results grid. The query window contains the following SQL code:

```
171  
172 ) AS l2_distance  
173 FROM Books  
174 WHERE embedding IS NOT NULL  
175 ORDER BY l2_distance ASC  
176 LIMIT 3;  
177
```

A tooltip above the query window reads: "Execute the selected portion of the script or everything, if there is no selection".

Below the query window is a "Result Grid" with the following data:

	book_id	title	author	genre	l2_distance
▶	1	The Martian	Andy Weir	Science Fiction	0
	8	Enders Game	Orson Scott Card	Science Fiction	0.0735
	2	Dune	Frank Herbert	Science Fiction	0.0866

At the bottom of the interface, there are tabs labeled "Books 6" and "Result 7".

SELECT

book_id,

title,

author,

genre,

ROUND(

(

CAST(


```

JSON_EXTRACT(embedding, '$[0]')
AS DECIMAL(10,6)
) * 0.90

+

CAST(
JSON_EXTRACT(embedding, '$[1]')
AS DECIMAL(10,6)
) * 0.80

+

CAST(
JSON_EXTRACT(embedding, '$[2]')
AS DECIMAL(10,6)
) * 0.20

)

/

(

SQRT(

POW(
CAST(
JSON_EXTRACT(embedding, '$[0]')
AS DECIMAL(10,6)
),

```

```

        2
    )

+

    POW(
        CAST(
            JSON_EXTRACT(embedding, '$[1]')
            AS DECIMAL(10,6)
        ),
        2
    )

+

    POW(
        CAST(
            JSON_EXTRACT(embedding, '$[2]')
            AS DECIMAL(10,6)
        ),
        2
    )

)

*

SQRT(

    POW(0.90, 2)

```

+

POW(0.80, 2)

+

POW(0.20, 2)

)

),

4

) AS cosine_similarity

FROM Books

WHERE embedding IS NOT NULL

ORDER BY cosine_similarity DESC

LIMIT 3;



The screenshot shows a database query result grid with the following data:

	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Andy Weir	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Below the table, there is a tab labeled "Result 8" and a button with an "x" icon. At the bottom, there is a label "Output" followed by a dotted line.